



A) Conjugate Linear Regression (posterior, predictive, evidence)

You observe

$$X = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{bmatrix}, \quad y = \begin{bmatrix} 0.2 \\ 1.1 \\ 1.9 \end{bmatrix}.$$

Model:

$$y | w \sim \mathcal{N}(Xw, \sigma^2 I), \quad \text{prior } w \sim \mathcal{N}(0, \alpha^{-1} I_2).$$

Use: $\sigma^2 = 0.10, \alpha = 4$.

1. Derive $p(w | y, X)$ and compute its mean and covariance.
2. Compute $p(y^* | x^*, y, X)$ at $x^* = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$.
3. Compute $\log p(y | X, \sigma^2, \alpha)$ (the evidence).
4. Compute w_{ML} and w_{MAP} ; interpret the shrinkage.

Rules:

$$p(z) = \mathcal{N}(z | \mu_z, \Sigma_z)$$

$$p(y | z) = \mathcal{N}(y | Wz + b, \Sigma_y)$$

$$\Rightarrow p(z | y) = \mathcal{N}(y | \mu_{z|y}, \Sigma_{z|y})$$

$$\mu_{z|y} = \Sigma_{z|y} \left[W^T \Sigma_y^{-1} (y - b) + \Sigma_z^{-1} \mu \right]$$

$$\Sigma_{z|y} = \Sigma_z^{-1} + W^T \Sigma_y^{-1} W$$

1) Derive $p(w | y, X)$ and compute μ and σ^2 .

- we have 0 prior $w \sim \mathcal{N}(0, \alpha^{-1} I_2)$
- 0 likelihood $y | w \sim \mathcal{N}(Xw, \sigma^2 I)$

$$S = \Sigma_{w|y} = \alpha I_2 + X^T \cdot \frac{1}{\sigma^2} I X$$

$$m = \mu_{w|y} = \left[\alpha I_2 + X^T \cdot \frac{1}{\sigma^2} I X \right]^{-1} \left[X^T \frac{1}{\sigma^2} I (y) \right]$$

$$= \Sigma_{w|y} \left[X^T \frac{1}{\sigma^2} I (y) \right]$$

$$\therefore p(w | y, X) = \mathcal{N} \left(\Sigma_{w|y} \left[X^T \frac{1}{\sigma^2} I (y) \right], \alpha I_2 + X^T \cdot \frac{1}{\sigma^2} I X \right)$$

$$= \mathcal{N} \left(\begin{bmatrix} 0.27 \\ 0.75 \end{bmatrix}, \begin{bmatrix} 0.05 & -0.03 \\ -0.03 & 0.03 \end{bmatrix} \right)$$

- using the class, we find: $m = \begin{bmatrix} 0.27 \\ 0.75 \end{bmatrix}$ $S = \begin{bmatrix} 0.05 & -0.03 \\ -0.03 & 0.03 \end{bmatrix}$

2) Compute $p(y^* | y, x^* = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, X) = \mathcal{N}(y^* | x^{*T} \mu_{post}, x^{*T} \Sigma_{post} x^* + \sigma^2)$

m = posterior mean
 S = posterior variance

- we use $p(y^* | y, x^* = \begin{bmatrix} 1 \\ 1 \end{bmatrix}) = \int p(y^* | w) p(w | y) dw$

$$= \mathcal{N}(y^* | m^T \phi^*, \phi^{*T} S \phi^* + \sigma^2)$$

$$m^T \phi^* = \begin{bmatrix} 0.27 & 0.75 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = 1.02$$

$$\phi^{*T} S \phi^* + \sigma^2 = \begin{bmatrix} 1 & 1 \end{bmatrix} \begin{bmatrix} 0.05 & -0.03 \\ -0.03 & 0.03 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} + 0.10$$

$$= 0.02 + 0.10$$

$$= 0.12$$

$$\therefore \mathcal{N}(y^* | 1.02, 0.12)$$

3) Log evidence

$$\log p(y | X, \sigma^2, \alpha) = -\frac{N}{2} \log(2\pi) - \frac{1}{2} \log |\Sigma_y| - \frac{1}{2} y^T \Sigma_y^{-1} y$$

$\hookrightarrow \Sigma_y = \sigma^2 I + \frac{1}{\alpha} X X^T$, alternatively using conjugate prior like

$$\log(y | X, \sigma^2, \alpha) = \mathcal{N}(y | 0, \sigma^2 I + \frac{1}{\alpha} X X^T)$$

4) $W_{MLE} \wedge W_{MAP}$
 $W_{MLE} = (\Phi^T \Phi)^{-1} \Phi^T y$
design matrix, but in our case just X

- W_{MAP} includes shrinkage term αI which pulls the estimate to 0.

$$W_{MAP} = \left(\alpha I + \frac{1}{\sigma^2} \Phi^T \Phi \right)^{-1} \cdot \left(\frac{1}{\sigma^2} \Phi^T y \right) = m \quad (\text{model is just posterior mean})$$

Gaussian processes

1.1 Form K and K_y

○ K is just matrix we found it by plugging our training inputs into $K(x, x')$
 $\therefore f \sim \mathcal{N}(f|0, K(x, x'))$
"K"

□ K_y is a marginal of y , found by marginalizing out the function values from:

$$p(y|0, C) = p(y|f) p(f) = \mathcal{N}(y|f, \beta^{-1}) \mathcal{N}(f|0, K)$$

$$C = K + \phi^{-1} I = K + \sigma^2 I$$

1.2 Compute posterior mean m_* , variance S_* at $x_* = \{0.25, 0.75, 1.25\}$

$$m_* = h_*^T K_y^{-1} y$$

$$S_* = \sigma^2 = h_{**} - h_*^T K_y^{-1} h_*$$

1.3 Compute $\log p(y|X, \theta)$

I compute the evidence, by using the provided class

1.4 Halving the lengthscale $\ell \rightarrow 0.5 \cdot \ell = 0.3$

- in GP w/ RBF kernel $K(x, x') = \sigma^2 \exp\left(-\frac{(x-x')^2}{2\ell^2}\right)$

ℓ controls how quickly the function
 small ℓ : more local influence $\Rightarrow$ functions vary rapidly

big ℓ : functions are smoother and more correlated over longer ranges

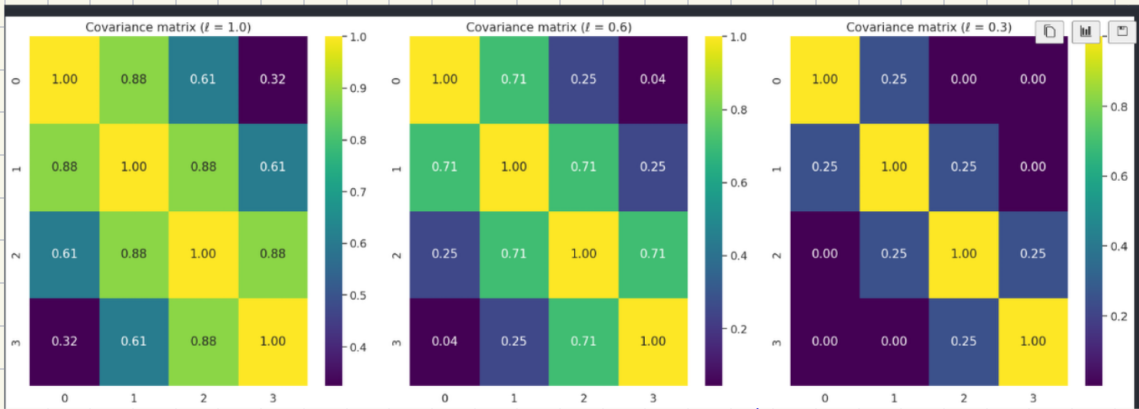
as we decrease ℓ , the kernel matrix K : as $\ell \rightarrow 0$, $k(x, x') \rightarrow 0$, except $x = x' \Rightarrow 1$.
 prior assumes that $f(x)$ & $f(x')$ are less correlated, so the GP becomes less confident in using info from neighboring points to predict at x_*

$\Rightarrow$ this also results in posterior variance increase

Post. variance at test point is: $\sigma_*^2 = h(x_*, x_*) - h_*^T K_y^{-1} h_*$

- when ℓ is small, K_* (cov. between test pt and training pt) becomes small

- the second term $K_*^T K_y^{-1} K_* \approx 0$, when $\ell \rightarrow 0$, therefore the variance becomes $\sigma_*^2 = 1 - \frac{K_*^T K_y^{-1} K_*}{\sigma^2} \approx 1 \Rightarrow$ large variance



$l=1$

- high values on the diagonals
- all pts. are strongly correlated
- GP prior believes functions are smooth (knowing one pt. gives me info about another)

$l=0.6$

- off diagonals decreased compared to $l=1$
- neighbour pts. are still correlated but the correlation slowly decays
- overall less smooth

$l=0.3$

- off diagonals are very small
- ONLY VERY CLOSE PTS. are correlated, else other pts. are treated as almost independent
- GP prior believes that the fun. can vary very rapidly.

1.5 Is the covariance kernel isotropic? stationary?

Stationary kernel = Covariance function ONLY depends on the difference between two points, ($y = x_n - x_m$)

Isometric kernel = Cov. function depends on the distance between two inputs $\|y\|_2 = \|x_n - x_m\|_2$.

Squared exp is both Isometric and Stationary

1) write analytical likelihood $p(y|w, \Phi, \sigma^2)$ & marginal likelihood $p(y, \sigma)$ in a closed form.

likelihood $p(y|w, \Phi, \sigma^2) = \mathcal{N}(\Phi w, \sigma^2 I)$

marginal likelihood $p(y) = p(y|w) p(w) = \mathcal{N}(y|0, \sigma^2 I + \alpha^{-1} \Phi \Phi^T)$